

# Where LLMs Fit Visual Repair

Focused repair pass for the AI family-tree visual. The folded topology map is now a clean coded taxonomy tree so learners can see that LLMs are one branch of generative AI inside the broader AI family.

## What Changed

### Visual Layout

Chose a clean vertical taxonomy tree because it was more readable than nested boxes at 320px and 390px.

### Hierarchy

The diagram now shows AI to Machine learning to Deep learning to Generative AI to LLMs, with diffusion and multimodal as neighboring branches.

### Labels

Removed compressed labels like ML, Gen AI, Multi, and Other DL. Classical ML remains as the allowed short label.

### Interaction

The tiny interaction is still tap-through, but now the hierarchy stays visible while the selected branch highlights and the explanation changes.

## Files Changed

- `src/components/DiagramKit.tsx`
- `src/components/VisualAids.tsx`
- `src/main.tsx`
- `src/styles/global.css`
- `docs/REVIEW_NOTES.md`
- `docs/design/DIAGRAM_KIT_V0_18.md`
- `docs/stage-audits/v0-17-before-morning/DIAGRAM_KIT_REFACTOR_V0_18.md`
- `docs/stage-audits/v0-17-before-morning/WHERE_LLMS_FIT_VISUAL_REPAIR_V0_18_1.md`
- `docs/stage-audits/v0-17-before-morning/screenshot-index.md`
- Report screenshots under `docs/stage-audits/v0-17-before-morning/screenshots/`

## Verification

typecheck passed

build passed

build:pages passed

audit:answers passed

browser QA passed

- `npm run typecheck`: passed.
- `npm run build`: passed with the existing Vite large-chunk warning.
- `npm run build:pages`: passed with the existing Vite large-chunk warning.
- `npm run audit:answers`: passed; 68 surfaces audited, 46 randomized, 22 fixed-order exclusions.
- Live browser QA: no console errors in a clean tab; no horizontal overflow at 320px; all ten interaction nodes visible and tappable.

## Known Issues

- The existing Vite large-chunk warning remains.
- The live mobile bottom navigation can cover the very bottom of long lesson content, but it does not obscure the repaired taxonomy diagram.
- No generated PNG was added for `ai-family-tree`; this card remains coded SVG/HTML.

## Next Three Recommended Steps

- Add a tiny regression check for `ai-family-tree` label presence, no process-number seals, and no horizontal overflow at 320px.
- Audit other coded visuals for similar “looks readable but teaches vaguely” problems.
- After this taxonomy repair is stable on iPhone, continue the planned Image 2 asset generation for cards that actually need textless art.

Screenshots

BEFORE MORNING2 / 39

1Before Morning

2Morning Commute

3Workday

4Decision Room

5The Day Repeats

6Twilight: The Wider Landscape

7Midnight Ledger

8New Dawn

### Where LLMs Fit

AI → machine learning → generative AI → LLMs

A large language model is one kind of generative AI system, built with deep-learning methods inside the broader field of artificial intelligence.

Previewing this card. Progress will not change.

VISUAL AID

What to picture

HomeJourneyPlayGlossaryBadge

Where LLMs Fit lesson top at 390px.

VISUAL AID

What to picture

```
graph TD; AI[AI] --> RuleBasedAI[Rule-based AI]; AI --> MachineLearning[Machine learning]; MachineLearning --> ClassicalML[Classical ML]; MachineLearning --> DeepLearning[Deep learning]; DeepLearning --> OtherDeepLearning[Other deep learning]; DeepLearning --> GenerativeAI[Generative AI]; GenerativeAI --> LLMs[LLMs]; GenerativeAI --> Diffusion[Diffusion]; GenerativeAI --> Multimodal[Multimodal];
```

#### AI Family Tree

**Where LLMs fit**

A clean taxonomy tree shows AI as the broad field, machine learning as one branch inside AI, and LLMs as one branch inside generative AI.

- AI** AI is the broad field.
- Machine learning** Machine learning systems learn patterns from data.
- Deep learning** Deep learning is a neural-network branch of machine learning.
- Generative AI** Generative AI creates new media, such as text, images, audio, code, or video.
- LLMs** LLMs generate language/code; diffusion and multimodal systems are neighboring generative AI branches.

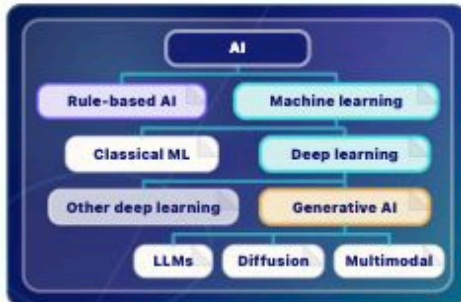
**Key takeaway:**  
An LLM is one kind of generative AI inside the

HomeJourneyPlayGlossaryBadge

AI family-tree visual at 390px.

## VISUAL AID

### What to picture

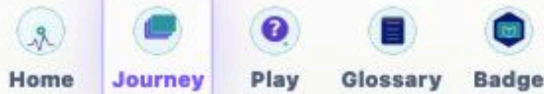


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- 5 **LLMs** LLMs generate language/code; diffusion and multimodal systems are neighboring generative AI



AI family-tree visual at 320px.

AI

Rule-based AI

Machine learning

Classical ML

Deep learning

Generative AI

LLMs

Diffusion

Multimodal

Other deep learning

#### Diffusion

Generative systems that often create images or other media by denoising patterns.

Reveal what each branch means in the AI family tree.

## CHECKPOINT



Home



Journey



Play



Glossary



Badge

Tap-through taxonomy interaction.

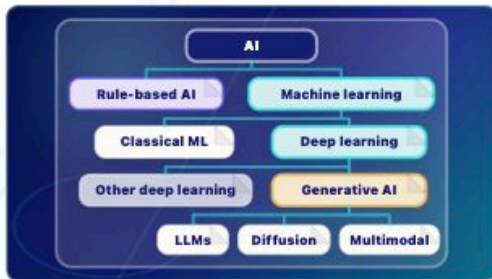
## AI-FAMILY-TREE

# AI Family Tree

Lesson: Where LLMs Fit

Pattern: aiTopology

Variant: origami-object



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### Key takeaway:

An LLM is one kind of generative AI inside the broader AI family.

Review route visual-aid card.